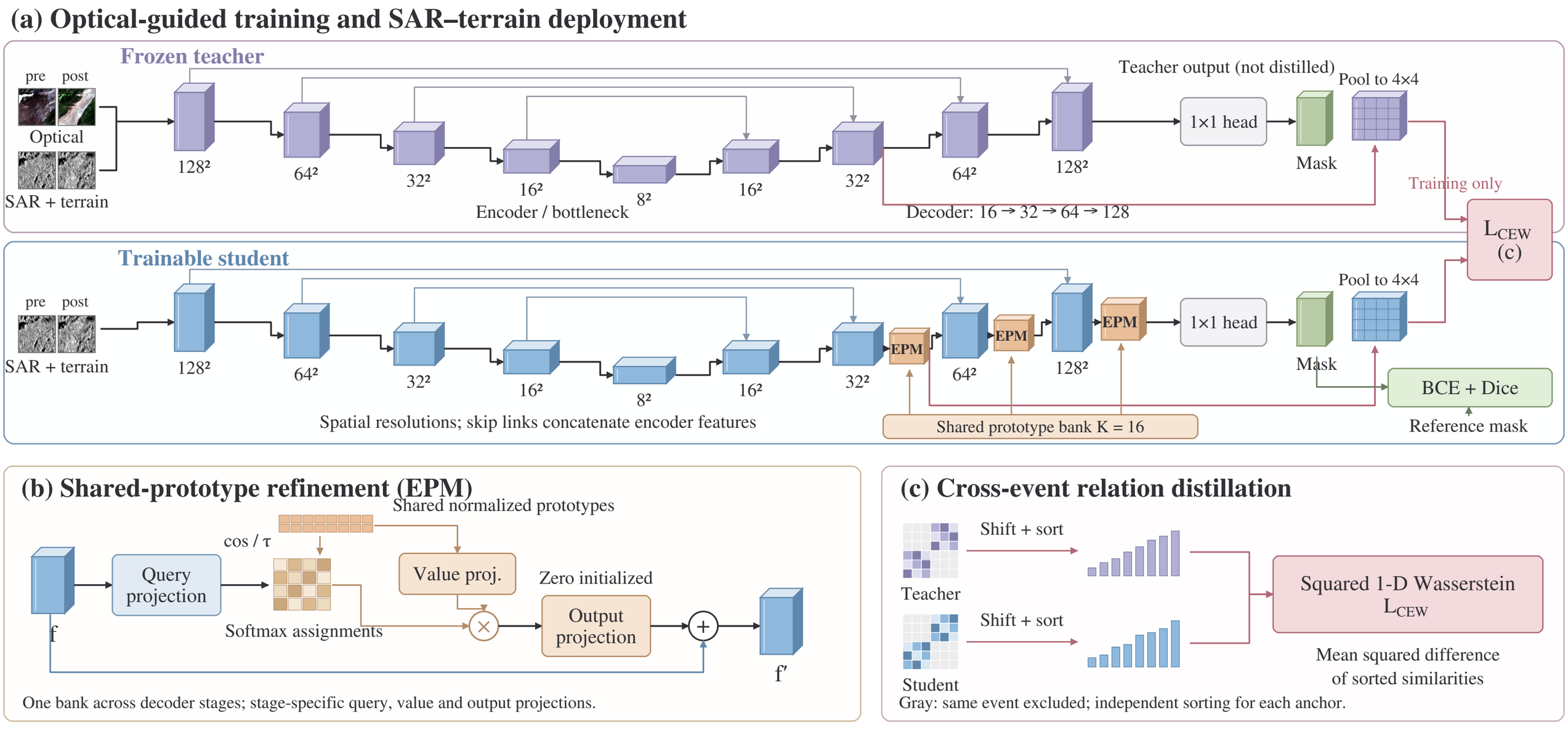

(a) Optical-guided training and SAR–terrain deployment
Frozen teacher
pre
post
Optical
SAR + terrain
128²
64²
32²
16²
8²
Encoder / bottleneck
Decoder: 16 → 32 → 64 → 128
Teacher output (not distilled)
1×1 head
Mask
Pool to 4×4
Training only
L_CEW (c)
Trainable student
EPM
Spatial resolutions; skip links concatenate encoder features
Shared prototype bank K = 16
BCE + Dice
Reference mask
(b) Shared-prototype refinement (EPM)
Shared normalized prototypes
cos / τ
Query projection
Value proj.
Zero initialized
Softmax assignments
Output projection
f
f′
One bank across decoder stages; stage-specific query, value and output projections.
(c) Cross-event relation distillation
Shift + sort
Teacher
Student
Squared 1-D Wasserstein L_CEW
Mean squared difference of sorted similarities
Gray: same event excluded; independent sorting for each anchor.